

Hormonal therapy for breast cancer

Hormonal therapy blocks the effects of oestrogen on breast cancer cells and reduces the risk of breast cancer coming back.

On this page

- [What is hormonal therapy?](#)
- [Having hormonal therapy for breast cancer](#)
- [Types of hormonal therapy](#)
- [Ovarian suppression or removing the ovaries \(ablation\)](#)
- [Side effects of hormonal therapy](#)
- [About our information](#)
- [How we can help](#)

What is hormonal therapy?

Hormones help control how cells grow and what they do in the body. The hormone oestrogen can encourage some breast cancers to grow. This may also happen with the hormone progesterone.

Hormonal therapy reduces the amount of oestrogen in the body or stops oestrogen attaching to cancer cells. It only works for breast cancer that is oestrogen receptor positive ([ER positive](#)). Hormonal therapy is also called endocrine therapy.

Your cancer doctor will advise you to have hormonal therapy to reduce the risk of breast cancer coming back. It also helps reduce the risk of getting a new breast cancer in your other breast.

You may be offered hormonal therapy before surgery. This aims to shrink the cancer. It may help you avoid a [mastectomy](#).

You usually have hormonal therapy drugs for a number of years. This may be for up to 10 years. You usually start taking them after surgery or chemotherapy.

The type of hormonal therapy you have depends on:

- whether you have been through the menopause
- the risk of the cancer coming back
- how the side effects may affect you.

We have more information about [different hormonal therapy drugs](#) and their side effects.

Having hormonal therapy for breast cancer

Hormonal therapy reduces the risk of [breast cancer](#) coming back. It is important to take it for as long as you have been prescribed it. Try to make taking it part of your daily routine so it becomes a habit.

Most people cope well with the side effects of hormonal therapy. Side effects may be more of a problem in the first few months, but they usually get better over time.

If the side effects do not improve or are difficult to cope with, talk to your breast cancer nurse or cancer doctor. They can give you drugs to help and can suggest ways of coping. Sometimes they may suggest changing the time you take the tablet. If you still have problems, they may suggest changing to a different type of hormonal therapy.

Types of hormonal therapy

Tamoxifen

[Tamoxifen](#) is an anti-oestrogen drug. It stops oestrogen attaching to breast cancer cells and making them grow.

You take tamoxifen once a day as a tablet.

You may have it if you have:

- not been through the menopause – you may take tamoxifen for a few years and then change to an aromatase inhibitor
- been through the menopause and have a low risk of the cancer coming back
- been through the menopause and are having problems with your bones – this is because tamoxifen does not cause bone thinning
- bad side effects with aromatase inhibitor drugs – you may be given tamoxifen instead.

If you are close to menopause when you start tamoxifen, you may change to an aromatase inhibitor after a few years. Your doctors may do blood tests to check your hormone levels first.

If you want to get pregnant, talk to your cancer doctor or nurse about the timing of this. You will need to have a break in your treatment, and you must stop taking tamoxifen for several months before trying to start a pregnancy (conceive). This is because tamoxifen can harm an unborn baby. Your doctor or nurse can give you more information about getting pregnant after breast cancer.

Aromatase inhibitors (AIs)

Aromatase inhibitors (AIs) are the main hormonal therapy drugs used if you have been through the menopause. They stop oestrogen being made in the fatty tissue, muscles and in the glands above the kidneys (adrenal glands). Your cancer doctor may prescribe an AI such as [anastrozole](#), [letrozole](#) or [exemestane](#). You take AIs daily as a tablet.

AIs may sometimes be given if you have not been through the menopause. But they can only be given with drugs to stop the ovaries working, or if the ovaries have been removed.

Your bone health

If you take AIs over a long time, they can cause bone thinning. This is called osteoporosis. You may have a scan to check your bone health (density). This is called a DEXA scan. Having your ovaries removed and ovarian suppression can also cause osteoporosis.

You may already be having a drug called a [bisphosphonate](#). These are given to help reduce the risk of cancer spreading to the bones. But they are also used to help protect from osteoporosis.

If you are at risk of osteoporosis and are not already having bisphosphonates, your cancer doctor may prescribe them to help protect your [bones](#).

Your doctor will usually check your calcium and vitamin D levels. If these are low, the doctor will advise you to take supplements to help strengthen your bones. There are things you can do to look after your bones, including eating healthily and doing regular weight-bearing exercise such as [walking](#).

Ovarian suppression or removing the ovaries (ablation)

If you have not been through the menopause, your cancer doctor may advise having 1 of these treatments with another hormonal therapy:

Ovarian suppression



This stops the ovaries making oestrogen and causes a temporary menopause.

Removing the ovaries (ablation)



This causes a permanent menopause.

These treatments lower oestrogen levels. This reduces the risk of breast cancer coming back. But they can also affect your bone health. You usually have a DEXA scan to check your bone health (density). Your doctor will advise you on how to [look after your bones](#).

Drugs to stop the ovaries making oestrogen (ovarian suppression)

[Goserelin \(Zoladex®\)](#) and [leuprorelin \(Prostap®\)](#) are drugs that stop the ovaries making oestrogen. This is called ovarian suppression. It causes a temporary menopause. This means you will not get a period while you are taking them.

These types of drugs stop the pituitary gland in the brain from sending messages to the ovaries to produce oestrogen. This causes side effects similar to [menopausal symptoms](#).

You have goserelin as a monthly injection under the skin of the tummy (abdomen). You usually have leuprorelin as an injection every 3 months. You usually have 1 of these drugs along with tamoxifen or an AI.

When you finish treatment, the ovaries usually start to produce oestrogen again. This means your periods will come back. But this may not happen if you were close to your natural menopause when you started treatment.

Removing the ovaries (ovarian ablation)

Removing the ovaries is called ovarian ablation. It may be an option if you do not want to have the injections. You will usually have tamoxifen or an AI as well.

Ovarian ablation involves a small operation. It is done using keyhole surgery. The surgeon makes a small cut in the tummy area. They then insert a thin tube through the cut. The tube is called a laparoscope. It has a tiny light and camera on the end and a small surgical instrument attached. The surgeon uses the laparoscope to remove the ovaries through the cut. You usually recover quickly from this operation, but you may need a short stay in hospital.

If you have ovarian ablation, your periods will stop straight away and you will start the menopause. Your periods will not come back. This means you will no longer be able to get pregnant. This can be hard to cope with. But there is lots of [support](#) available.

Hormonal therapy before the menopause

If you have not been through the menopause, your cancer doctor may recommend 1 of the following:

- [tamoxifen](#)
- [goserelin](#) or [leuprorelin](#) – with tamoxifen or an AI
- surgery to remove the ovaries – on its own or with tamoxifen or an AI.

Hormonal therapy after the menopause

If you have been through the menopause, your doctor may prescribe 1 of the following:

- an AI such as [anastrozole](#), [letrozole](#), [exemestane](#)
- [tamoxifen](#) and an AI (one type is given after the other)
- tamoxifen on its own.

Side effects of hormonal therapy

Different hormonal therapy drugs have different side effects. Some of the main side effects include:

- hot flushes and sweats
- weight gain
- joint and muscle pain
- tiredness
- sexual effects, including vaginal dryness.

Some types of hormonal therapy can slightly increase the risk of developing a blood clot. Tamoxifen may also slightly increase the risk of womb changes and womb cancer. Tell your cancer doctor if you have any vaginal discharge or bleeding.

We have more information about the side effects of [different hormonal therapy drugs](#).

About our information

References



Below is a sample of the sources used in our breast cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

ESMO. Early breast cancer clinical practice guidelines for diagnosis, treatment and follow-up. 2019, Vol 30, pp1192–1220. Available from: <https://www.esmo.org/guidelines/guidelines-by-topic/breast-cancer/early-breast-cancer> [accessed 2023].

National Institute for Health and Care Excellence (NICE). Early and locally advanced breast cancer: diagnosis and management. 2018. Updated 2023. Available from: <https://www.nice.org.uk/guidance/ng101> [accessed 2023].

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by [Dr Rebecca Roylance](#), Consultant Medical Oncologist and [Professor Mike Dixon](#), Professor of Surgery and Consultant Breast Surgeon.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
-

- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 October 2023 | Next review: 01 October 2026

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help